

**ADMINISTRATIVE INFORMATION****FMI No: 22904/22905****FMI Title: NGPDU Series Overheat Protection Replacement**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☐ Mandatory Safety ☒ Mandatory ☐ Mandatory on Request	GEMS-AM: Code AM GEMS-E: Code E GEMS-A: Code A	Issue Date: Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☒ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☐ No Parts are required to complete this FMI.
☐ Batch with	To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5148718****Do NOT leave unsecured on site.**



Completion Certification Report

Mail To

GEMS - Americas

GE Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - E / ESC
FMI Admin. 322
BP34
F 78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 22904/22905 FMI Title: NGPDU Series Overheat Protection Replacement

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

- 1.) Modification Complete or previously modified.
- 2.) FMI not required per FMI instruction.
- 3.) Not modified. Unit in storage. FMI kit returned to stock.
- 4.) Modification refused or equipment altered. Customer signature and title required.
- 5.) Unit location unknown or unit not in area. Product Locator notified.

Customer Refusal or Equipment Altered - Modification Pending.

Installer
Comments

EFFECTIVITY:

Please contact the local FMI administrator to obtain the actual customer sites.

FMI No.	FMI Kit P/N	NGPDU Type	NGPDU Part No.
FMI22904	5148582	NGPDU NGPDU-3	2326492 2326492-3
FMI22905	5151225	NGPDU-2	2326492-2



NOTICE Forward production has started to implement this change starting from Serial number E4YP05408(NG-4).

PURPOSE:

If the IGBT QB1 on the NGPDU Series control board failed, the discharge resistors RB12 through RB17 will be exposed to HVDC 700V, the result is that these resistors will get very hot and scorch the control PCB and the Lexan top cover, smoke will be emitted.

This FMI is to add thermal alarm reset circuitry to prevent the RB12 through RB17 from being exposed to 700Vdc when IGBT Q1 failed and thus avoid smoking of NGPDU Series.

RELATED FMI'S:

None

MANPOWER REQUIREMENTS:

One field engineer, one hour labor plus travel

SPECIAL TOOLS & TEST EQUIPMENT

None

FURNISHED MATERIALS

FMI22905 kit (5151225) consists of the following: (for 2326492-2)

ITEM	PART NO.	DESCRIPTION	QTY.	COMMENT
1	5146322	Protect Bd Top Steel	1	
2	5146324	Temperature Sensor	1	
3	5150260	Signal Cable	1	
4	N9306HR	SCREW TP 3X6 FE ZN	2	
5	U0202AC	Clamp	100	
6	5151248	Jumper	1	
7	5148718	NGPDU series overheat protection replacement	1	this manual

FMI22904 kit (5148582) consists of the following: (for 2326492 and 2326492-3)

ITEM	PART NO.	DESCRIPTION	QTY.	COMMENT
1	5146322	Protect Bd Top Steel	1	
2	5146324	Temperature Sensor	1	
3	5147886	Signal Cable	1	
4	N9306HR	SCREW TP 3X6 FE ZN	2	
5	U0202AC	Clamp	100	
6	5151248	Jumper	1	
7	5148718	NGPDU series overheat protection replacement	1	this manual

PROCEDURE



WARNING

For systems powered from a Power Distribution Box which has a lockout capability, Turn OFF and lock out the system power at the main input before the replacement procedure.



CAUTION

ELECTRICAL HAZARD!

Please be assured that you have understood all the safety guidelines that you have to follow during your work.

Please Turn OFF the Main Power Distribution Supply to CT system and Switch OFF the CB1 during your FMI work.

1 PREPARATION

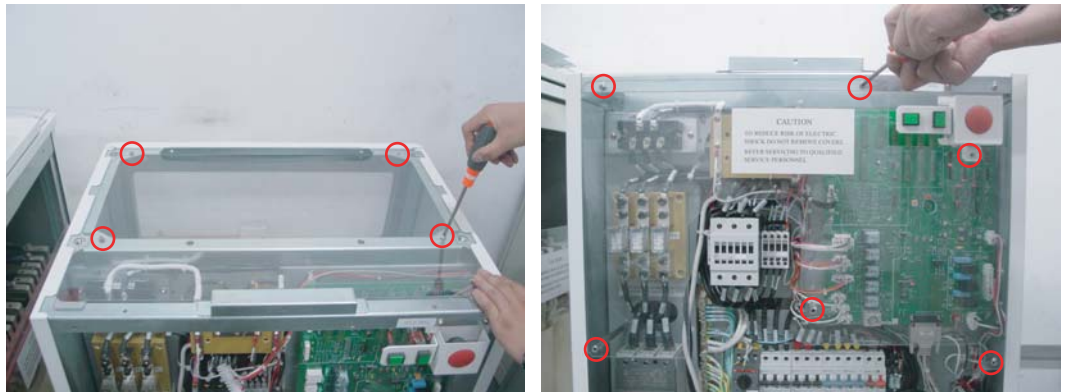
Understand and follow the Safety Guidelines Manual.

- 1.) Switch OFF the OC main power.
- 2.) Switch OFF the breaker in the Power Distribution Box (hospital supply).

Note: If there is UPS, Switch OFF the UPS output.

- 3.) Remove the Top and Front Covers from the NGPDU.
- 4.) Switch OFF the CB1 at the front side of the NGPDU.
- 5.) Remove the Top and Front Plastic covers. (See Illustration [Figure 1](#))

Figure 1 - Remove the Plastic Covers



2 Overheat Protection Installation

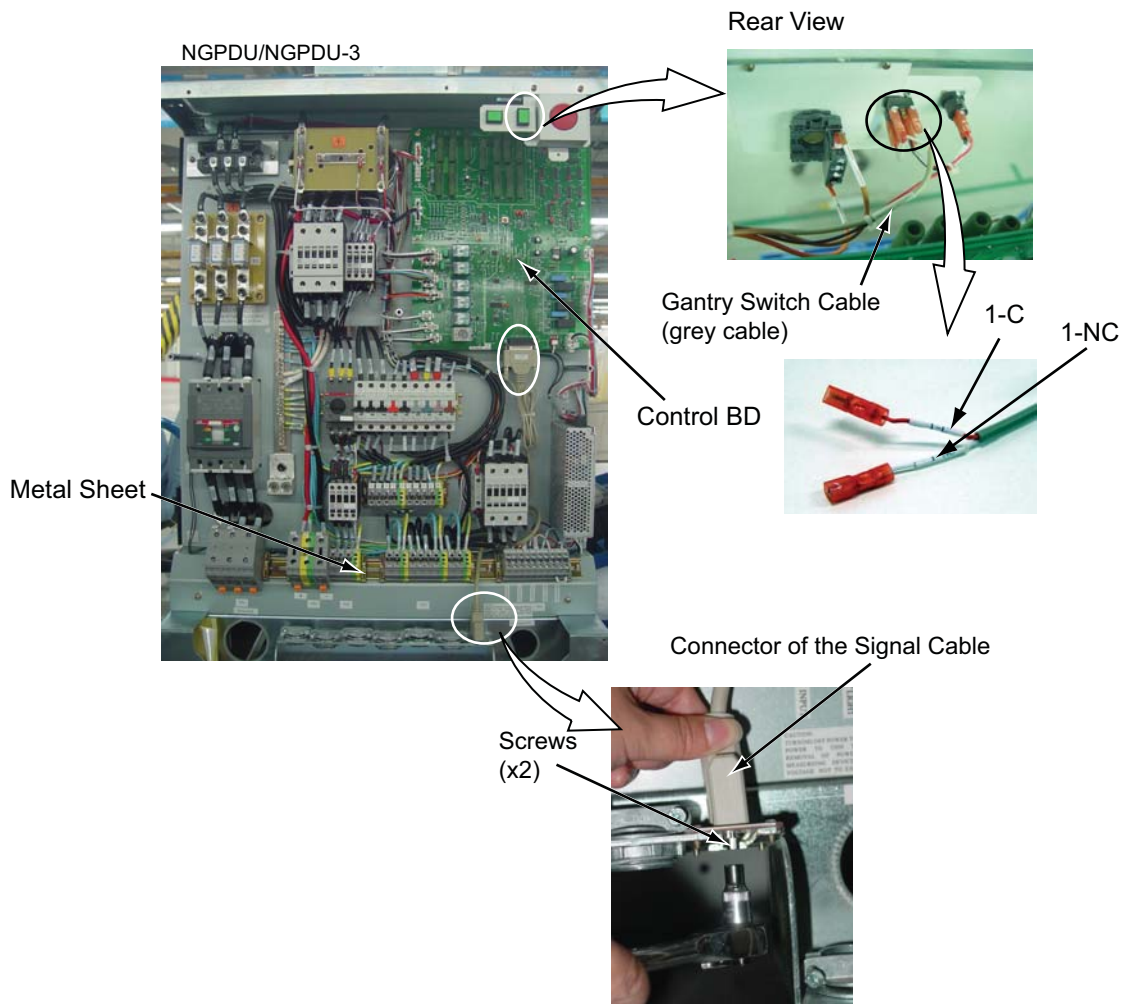
2.1 For FMI22904 (NGPDU/NGPDU-3)

- 1.) Remove the old signal cable. (connectors within white circles)

Note: Skip Step i for NGPDU/NGPDU-3 without Gantry Switch and go to Step ii.

- i.) Carefully remove the connectors of signal cable from the Gantry Switch, avoid causing damage to the cable or the Gantry Switch. (See [Figure 2](#))

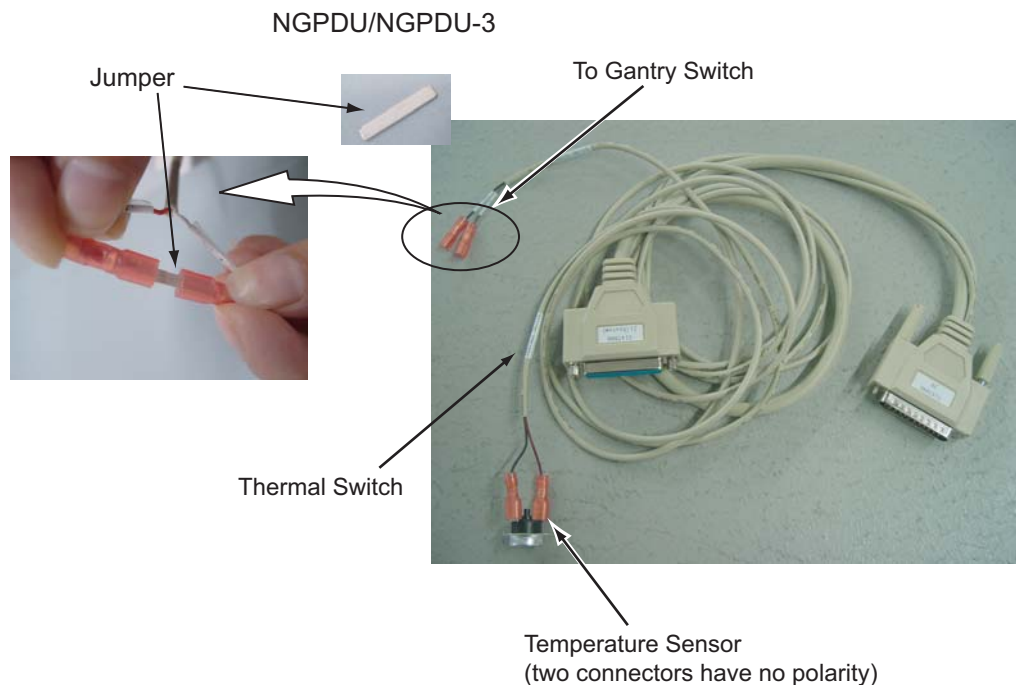
Figure 2 - Remove the Signal Cable



- ii.) Remove the connector of signal cable from the PDU Control Board. (See [Figure 2](#))
- iii.) Remove the connector of signal cable from the metal sheet by unscrewing its two screws with 5mm socket wrench. (See [Figure 2](#))

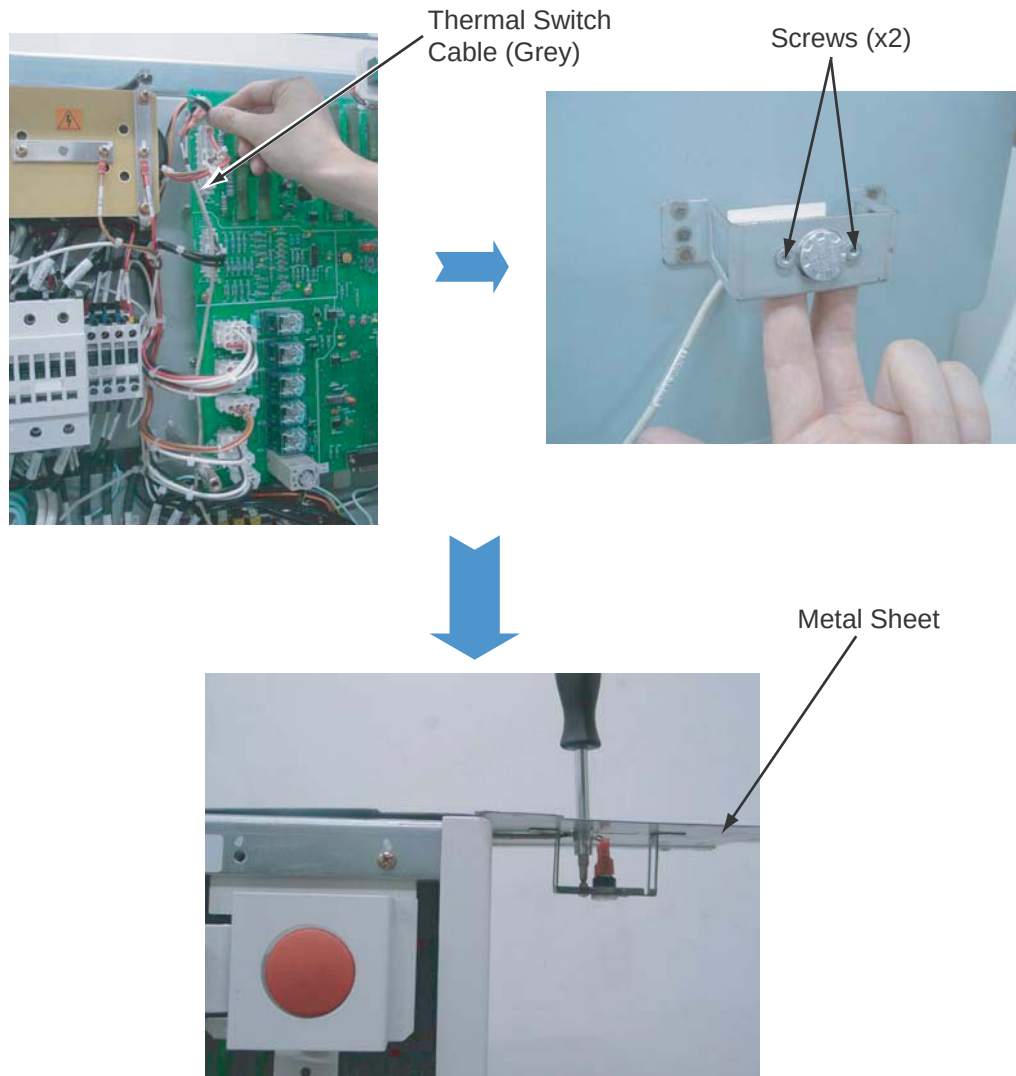
- 2.) Connect the Temperature Sensor to the end of the thermal switch of the new signal cable.
- 3.) For NGPDU/NGPDU-3 without the Gantry Switch, short the two connectors together with the jumper (5151248) provided with the FMI kit (5148582).

Figure 3 - Install the Temperature Sensor



- 4.) Pull the Thermal Switch Cable to go through the wires from the bottom up. (See [Figure 4](#))
- 5.) Place the Temperature Sensor on the bracket of the metal sheet (5146322). (See [Figure 4](#))
Fasten two screws on the Temperature Sensor.

Figure 4 - Route the Signal Cable and fix the Temperature Sensor on the metal sheet

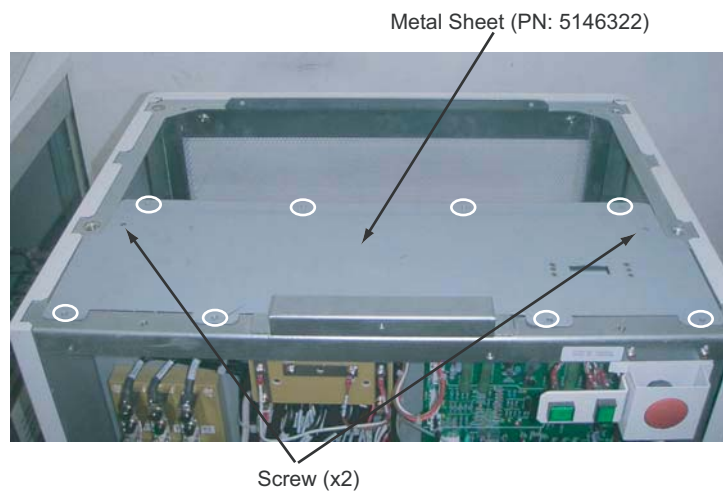
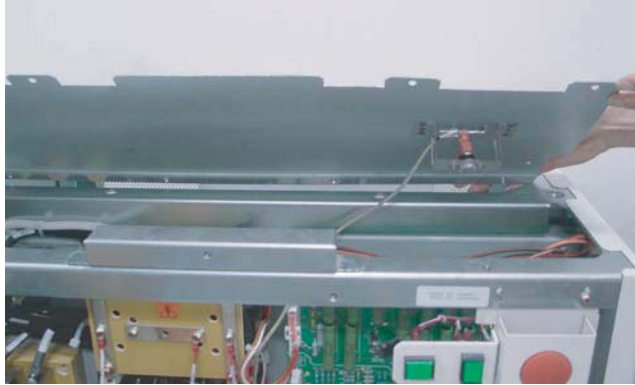


- 6.) Put the sheet metal (5146322) onto the top of the NGPDU/NGPDU-3. (See [Figure 5](#))



NOTICE Do not install the two screws (shown as [Figure 5](#)) at this time.

Figure 5 - Intall the metal sheet



Note: For this FMI, eight holes (white circles in above figure) are unused.

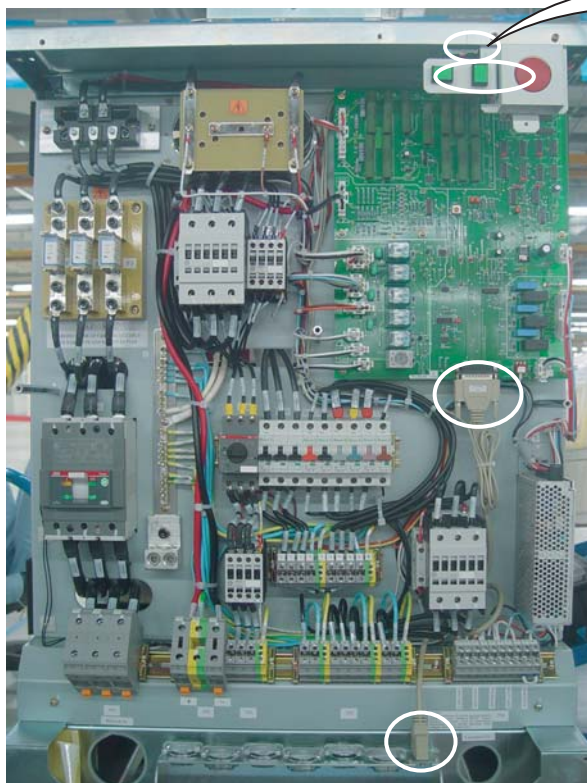
- 7.) According to [Step 1](#) process of remove the old Signal Cable in its reverse order to connect the new Signal Cable, then use tie-wraps to hold the cable. (See Illustration [Figure 6](#))

**NOTICE**

For NGPDU/NGPDU-3 without the Gantry Switch, pay attention to bind the Gantry Switch Cable.

Figure 6 - Install the Signal Cable

NGPDU/NGPDU-3



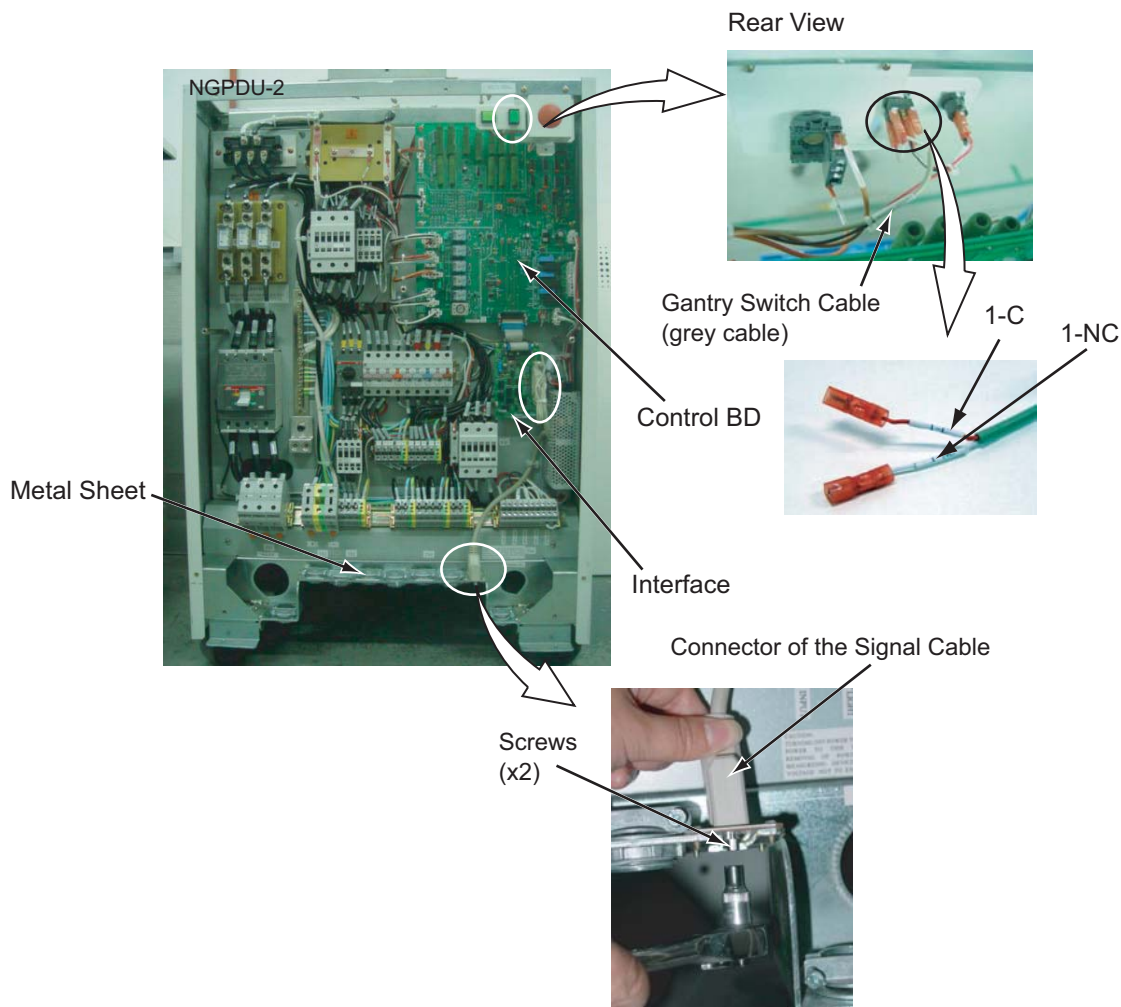
2.2 For FMI22905 (NGPDU-2)

- 1.) Remove the old signal cable. (connectors within white circles)

Note: Skip Step i for NGPDU-2 without Gantry Switch and go to Step ii.

- i.) Carefully remove the connectors of signal cable from the Gantry Switch, avoid causing damage to the cable or the Gantry Switch. (See [Figure 7](#))

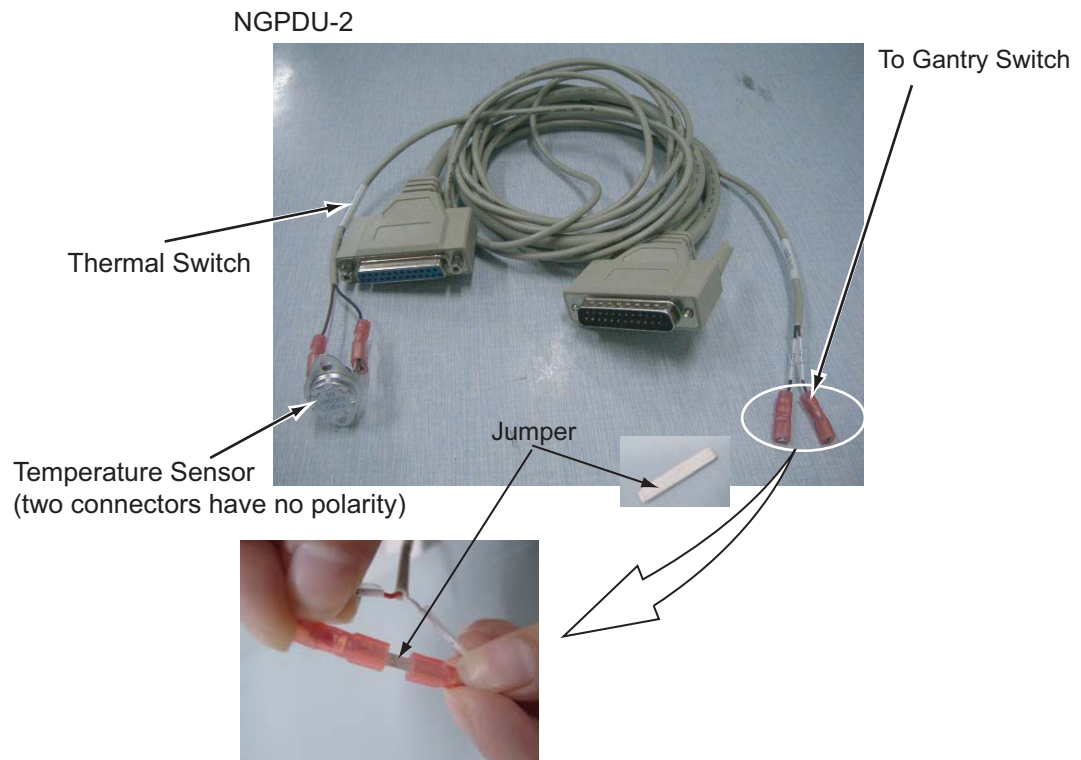
Figure 7 - Remove the Signal Cable



- ii.) Remove the connector of signal cable from the Interface. (See [Figure 7](#))
 - iii.) Remove the connector of signal cable from the metal sheet by unscrewing its two screws with 5mm socket wrench. (See [Figure 7](#))

- 2.) Connect the Temperature Sensor to the end of the thermal switch of the new signal cable.
- 3.) For NGPDU-2 without the Gantry Switch, short the two connectors with the jumper (5151248) provided with the FMI kit (5151225).

Figure 8 - Install the Temperature Sensor



- 4.) Pull the Thermal Switch Cable to go through the wires from the bottom up. (See [Figure 4](#)).
- 5.) Place the Temperature Sensor on the bracket of the metal sheet (5146322). (See [Figure 4](#)) Fasten two screws on the Temperature Sensor.
- 6.) Put the metal sheet (5146322) onto the top of the NGPDU-2. (See Illustration [Figure 5](#))



NOTICE

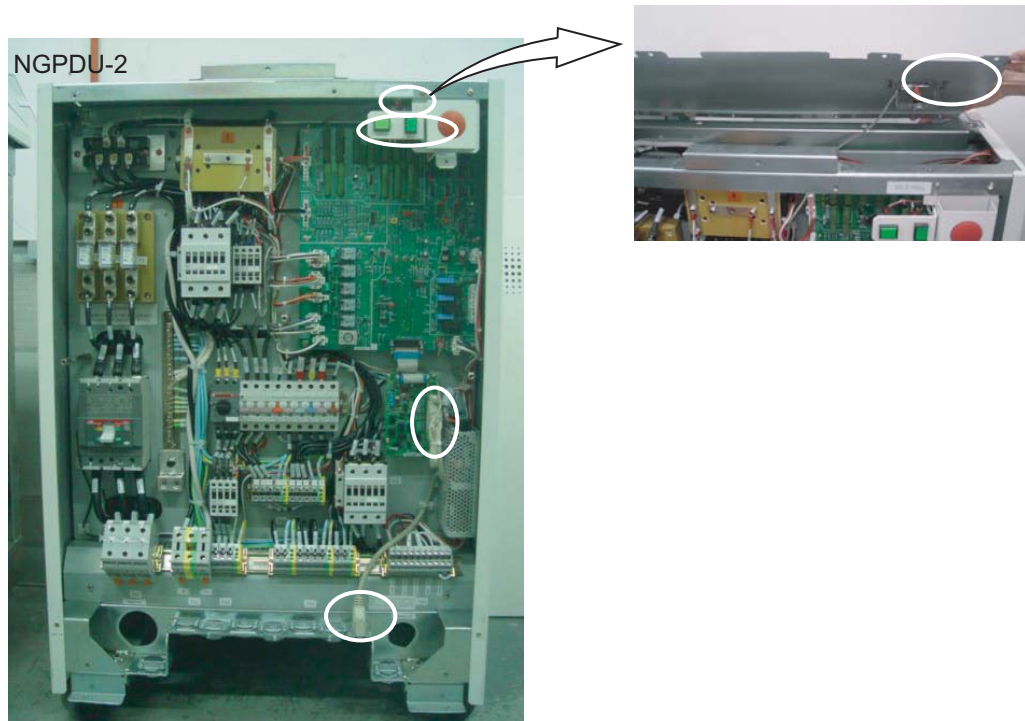
Do not install the two screws (shown as [Figure 5](#)) at this time.

- 7.) According to [Step 1](#) process of remove the old Signal Cable in its reverse order to connect the new Signal Cable, then use tie-wraps to hold the cable. (See Illustration [Figure 9](#))



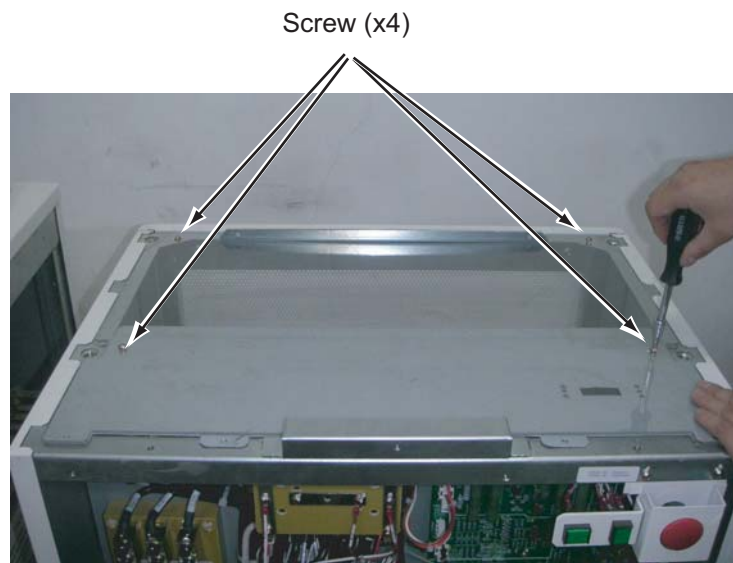
NOTICE For NGPDU-2 without the Gantry Switch, pay attention to bind the Gantry Switch Cable.

Figure 9 - Install the Signal Cable



3 Completing installation.

- 1.) Install the Top Plastic Cover.



- 2.) Install the Front Plastic Cover.
- 3.) Switch ON the CB1 at the front side of the NGPDU.
- 4.) Power on the system and verify it functions normally.
- 5.) Install the Front and Top Covers of the NGPDU.
- 6.) End.